Contribution Report

MLE-AI-201 Machine Learning — Final Project, team `ML-Team-08`

Repository: https://github.com/team-M503-SWE-06/ml_final_project

Everything below can be checked against the Git history of `main` and the merged pull requests.

Team

Member (Surname, Name)	GitHub handle	Main responsibilities
Aliyeva, Nurana	`Nurana100`	Experiments: metrics module, tree and SVM sweeps, benchmark vs scikit-learn
Garibli, Sanan	`sanan-garibli`	Data: repo skeleton, cleaning, features, frozen split, tier label, EDA, `run_all`
Jamalov, Humbat	`Camalzadeh`	Decision tree: from-scratch tree, its tests, permutation importance, tree hyperparameter search
Nematzada, Farah	`f4r4hh`	Error analysis and its `run_all` step; final report and slides; test-suite and full-pipeline verification runs
Shukurov, Elsen	`elsensukurov`	SVM: Pegasos SVM, random Fourier features, SVM tests and analysis

Per-member summary

Aliyeva, Nurana — Wrote the NumPy metrics module `src/evaluate.py` with tests that cross-check every metric against scikit-learn (`tests/test_evaluate.py`), together with the validation-only sweeps `src/experiments/tree_sweeps.py` and `src/experiments/svm_sweeps.py` (PR #4; the earlier PRs #2 and #3 were folded into it). Built `src/experiments/benchmark.py` (PR #5: 5-fold CV mean ± std, timings, a single test-set evaluation) and reworked it in PR #14 (fixed configs, optional SVM, tree-structure comparison with scikit-learn, sklearn baselines inside `Pipeline`s, smearing-corrected AZN metrics). Gave 5 PR reviews.

Garibli, Sanan — Created the repo skeleton (first commit) and `src/data_prep.py` in PR #1: cleaning of the raw bina.az dump, feature construction, the frozen stratified train/val/test split and the premium/standard tier label, plus the EDA figures in `src/eda.py`. In PR #16 he made `src/run_all.py` run every experiment end to end and fixed the `numpy` pin in `requirements.txt`. Gave 17 PR reviews, the most on the team.

Jamalov, Humbat — Implemented `src/decision_tree.py` from scratch in PR #6 (Gini/entropy/MSE, vectorised split search, four stopping rules, `predict_proba`, feature importances, input validation, growth with an explicit stack) and its test suite `tests/test_decision_tree.py` in PR #7 (node-by-node match with scikit-learn on fully determined examples, stopping rules, validation; 24 tests at merge). Added permutation importance with one-hot/missing-flag groups in `src/importance.py` with `tests/test_importance.py` (PR #8), and ran the 5-fold CV hyperparameter search whose tree configs are used in `benchmark.py` (`TREE_REG_PARAMS`, `TREE_CLS_PARAMS`). Reviewed #5 and #10 (4 reviews in total)

Nematzada, Farah — Wrote `src/error_analysis.py` in PR #18: test-set error analysis of the decision trees by price band, by district and by distance to the tier threshold, and hooked it into `src/run_all.py` (section 7b), so that the one command also reproduces the error tables and figure. Its tree settings are the final ones from `benchmark.py`, so it analyses the same trees as the results table. Wrote the report and finalized the slides. Outside Git (not visible in the history), she ran the test suite (53 tests passed) and the full `python -m src.run_all` pipeline on her own machine and checked that the error-analysis step reproduces the benchmark numbers (test RMSE on log price 0.332, MAE 0.193).

Shukurov, Elsen — Implemented the Pegasos SVM in `src/svm.py` (solver, per-epoch objective, margin and support vectors, random Fourier features, mini-batches and iterate averaging) with `tests/test_svm.py` and the analysis script `src/experiments/svm_analysis.py`, merged as PR #15. The work was first split into PRs #10–#13, which were closed in favour of #15. Gave 5 PR reviews.

Rough effort split

The approximate percentages were agreed by the team.

Member	Approx. %	Merged PRs	Non-merge commits on `main`	Lines added	Reviews
Aliyeva, Nurana	∼20%	#4, #5, #14	8 (+3, 2nd e-mail)	∼675	5
Garibli, Sanan	∼20%	#1, #16	5	∼1363	17
Jamalov, Humbat	∼20%	#6, #7, #8	20	∼752	4
Nematzada, Farah	∼20%	#18, 19, 20	2	∼161	1
Shukurov, Elsen	∼20%	#15	11	∼780	5